

From point through density valuation to individual risk assessment in the discounted cash flows method.

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Motivation

A POINT ESTIMATE CONCEALS THE UNCERTAINTY THAT MATTERS

- The discounted-cash-flow method delivers a single number, $V = \sum_t \mathbb{E}[CF_t] / (1 + r)^t$, from inputs that are themselves uncertain: projected cash flows and the discount rate.
- Sensitivity tables and scenario grids, the standard practice, probe the input space pointwise and carry no probability content.
- The paper reformulates the valuation to deliver the full distribution of outcomes implied by explicit distributions on the inputs.

Density valuation

STOCHASTIC DCF, MONTE CARLO AND KERNEL DENSITY ESTIMATION

Cash flows are represented by revenue, cost, scaled and fixed components,

$$CF(t) = \sum_{i=1}^N R_i(t) - \sum_{i=1}^M C_i(t) + \sum_{i=1}^K S_i((R_n), (C_m), (Y_e), t) + F,$$

with explicit stochastic processes on the drivers and on the discount rate, so the valuation V becomes a random variable.

- The valuation density is generated parsimoniously by Monte Carlo simulation and estimated by kernel density methods; a Cornish–Fisher expansion is implemented as the analytic alternative, capturing skewness and kurtosis beyond the Gaussian approximation.

| From density to individual risk assessment

DOWNSIDE MEASURES FOR HETEROGENEOUS USERS

- The paper selects and discusses density-based downside risk measures that appraisers and analysts can attach to a DCF valuation: quantile-based bounds on value at chosen confidence levels and probabilities of falling short of critical levels.
- The target audience is a heterogeneous (by risk appetite) group of final users: each reads the same valuation density through their own risk tolerance, without relying on a single aggregated point recommendation.
- Valuation and risk measurement collapse into one object: the same density answers the pricing and the risk question.

Implications

FOR VALUATION PRACTICE

- Reporting a valuation as a distribution disciplines the input assumptions: every scenario grid becomes a special case of an explicit probabilistic statement.
- Disagreements between valuations can be decomposed into disagreements about input distributions, which are auditable.
- The framework extends naturally to any present-value object with uncertain flows and rates, from equity valuation to project finance.

Thank you.

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